Assignment: Deterministic Decision Tree

1. Introduction

A decision tree is a model used in Artificial Intelligence to make decisions based on conditions. It has a tree-like structure where each node represents a decision and branches represent possible outcomes.

A **deterministic decision tree** always produces the same output for the same input, without any randomness.

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2. Objective

To design a deterministic decision tree to predict whether a student will pass or fail based on study hours and attendance.

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3. Problem Statement

Predict student performance (PASS/FAIL) using:

- Study Hours
- Attendance Percentage

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4. Decision Tree Diagram

```
              [Start]
                 |
         Study Hours >= 5?
            /      \
          Yes        No
          /           \
 Attendance >= 75%?       FAIL
    /    \
   Yes     No
   /         \
 PASS        FAIL
```

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5. Algorithm

1. Start
2. Check if Study Hours ≥ 5
   - If No → FAIL
3. If Yes, check Attendance ≥ 75%
   - If Yes → PASS

    - If No → FAIL

4. Stop

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6. Implementation (Python Code)

```
def decision_tree(study_hours, attendance):
    if study_hours >= 5:
        if attendance >= 75:
            return "PASS"
        else:
            return "FAIL"
    else:
        return "FAIL"

print(decision_tree(6, 80))
```

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7. Explanation

The model first checks study hours. If they are less than 5, the student fails. If sufficient, attendance is evaluated. Only students meeting both conditions pass.

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8. Advantages

- Simple and easy to understand
- Fast decision-making
- Predictable results

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9. Disadvantages

- Not suitable for complex problems
- Requires manual rules
- Less flexible than advanced AI models

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10. Conclusion

Deterministic decision trees provide a simple and reliable method for decision-making in basic AI applications.
